

# Project: Transportation Fleet Management System (TFMS)

## 1. Overview

The **Transportation Fleet Management System (TFMS)** is a web-based solution designed to manage and optimize fleet operations in the logistics domain. It supports vehicle tracking, trip scheduling, fuel management, maintenance tracking, and performance analysis. This system enhances operational efficiency, reduces costs, and ensures compliance with regulations.

The system is developed following the **MVC architecture** and is compatible with both **Java (Spring MVC)** and **.NET Core (ASP.NET Core MVC)** frameworks.

## 2. Core Modules Description

### 2.1 Vehicle Management

Tracks details of vehicles in the fleet, including registration, capacity, and status.

### 2.2 Trip Scheduling

Plans and manages trips, including route assignments, schedules, and driver allocations.

### 2.3 Fuel Management

Monitors and optimizes fuel consumption and costs.

### 2.4 Maintenance Tracking

Keeps track of vehicle maintenance schedules and history.

### 2.5 Performance Analysis

Generates reports and dashboards to analyze fleet performance and efficiency.

## 3. Module-Level Design

### 3.1 Vehicle Management Module

**Purpose:** Manages vehicle records and statuses.

- **Controller:**
  - VehicleController
    - addVehicle()
    - updateVehicle()
    - deleteVehicle()
    - viewVehicle()
- **Service:**
  - VehicleService

- Handles business logic related to vehicle management.
- **Model:**
  - **Vehicle Entity**
    - Attributes:
      - vehicleId (PK)
      - registrationNumber
      - capacity
      - status
      - lastServicedDate

### 3.2 Trip Scheduling Module

**Purpose:** Manages trips, including route and driver assignments.

- **Controller:**
  - TripController
    - addTrip()
    - updateTrip()
    - viewTrips()
- **Service:**
  - TripService
    - Contains logic for scheduling and route assignments.
- **Model:**
  - **Trip Entity**
    - Attributes:
      - tripId (PK)
      - vehicleId (FK)
      - driverId (FK)
      - startLocation
      - endLocation
      - startTime
      - endTime

### 3.3 Fuel Management Module

**Purpose:** Tracks and manages fuel usage and costs.

- **Controller:**
  - FuelController
    - addFuelRecord()
    - viewFuelUsage()
- **Service:**
  - FuelService
    - Handles logic for fuel tracking and cost analysis.
- **Model:**
  - **Fuel Entity**
    - Attributes:
      - fuelId (PK)
      - vehicleId (FK)

- date
- fuelQuantity
- cost

### 3.4 Maintenance Tracking Module

**Purpose:** Tracks and schedules vehicle maintenance.

- **Controller:**
  - MaintenanceController
    - scheduleMaintenance()
    - viewMaintenanceRecords()
- **Service:**
  - MaintenanceService
    - Manages maintenance schedules and history.
- **Model:**
  - **Maintenance Entity**
    - Attributes:
      - maintenancelId (PK)
      - vehicleId (FK)
      - description
      - scheduledDate
      - status

### 3.5 Performance Analysis Module

**Purpose:** Provides insights into fleet performance and operational efficiency.

- **Controller:**
  - PerformanceController
    - generateReport()
    - viewDashboard()
- **Service:**
  - PerformanceService
    - Generates reports and analytics.
- **Model:**
  - **Performance Entity**
    - Attributes:
      - performancelId (PK)
      - reportType
      - data
      - generatedOn

## 4. Database Schema

### Database Tables

1. Vehicle

```
CREATE TABLE Vehicle (  
    vehicleId INT AUTO_INCREMENT PRIMARY KEY,  
    registrationNumber VARCHAR(50),  
    capacity INT,  
    status VARCHAR(50),  
    lastServicedDate DATETIME  
);
```

## 2. **Trip**

```
CREATE TABLE Trip (  
    tripId INT AUTO_INCREMENT PRIMARY KEY,  
    vehicleId INT,  
    driverId INT,  
    startLocation VARCHAR(100),  
    endLocation VARCHAR(100),  
    startTime DATETIME,  
    endTime DATETIME,  
    FOREIGN KEY (vehicleId) REFERENCES Vehicle(vehicleId)  
);
```

## 3. **Fuel**

```
CREATE TABLE Fuel (  
    fuelId INT AUTO_INCREMENT PRIMARY KEY,  
    vehicleId INT,  
    date DATETIME,  
    fuelQuantity DECIMAL(10, 2),  
    cost DECIMAL(10, 2),  
    FOREIGN KEY (vehicleId) REFERENCES Vehicle(vehicleId)  
);
```

## 4. **Maintenance**

```
CREATE TABLE Maintenance (  
    maintenancelId INT AUTO_INCREMENT PRIMARY KEY,  
    vehicleId INT,  
    description VARCHAR(255),  
    scheduledDate DATETIME,  
    status VARCHAR(50),  
    FOREIGN KEY (vehicleId) REFERENCES Vehicle(vehicleId)  
);
```

## 5. **Performance**

```
CREATE TABLE Performance (  
    performanceId INT AUTO_INCREMENT PRIMARY KEY,  
    reportType VARCHAR(50),  
    data TEXT,  
    generatedOn DATETIME
```

);

## 5. Local Deployment Details

1. **Database Setup:** Install and configure **MySQL** or **SQL Server**, and execute the above script to initialize the database.
2. **Environment Setup:**
  - Java: Use **Maven** for dependency management and deploy the application on **Apache Tomcat**.
  - .NET Core: Use **Visual Studio** to build and deploy locally on IIS.
3. **Local Access URLs:**
  - Java: `http://localhost:8080/tfms`
  - .NET Core: `http://localhost:5000/tfms`

## 6. Conclusion

The **Transportation Fleet Management System (TFMS)** ensures efficient fleet operations, enabling seamless management of vehicles, trips, fuel, and maintenance. This scalable and modular design is tailored to enhance productivity and operational transparency in the logistics domain.